

CHRISTOPH LINSE

Cloud and Software Engineer



SKILLS

Kubernetes 5+ yrs

Openshift 5+ yrs

Containers 5+ yrs

Linux 5+ yrs

Java 5+ yrs

Golang 3+ yrs

Python 3+ yrs

CONTACT

📍 Wylerfeldstr. 14
3014 Bern
📞 +41 76 264 10 72
✉ christoph.linse@gmail.com



PROFILE

I am a cloud and software engineer who gained a lot of valuable experience around frontend- and backend development, as well as software provisioning and automation using infrastructure as code.

My enthusiasm for cloud native computing has lasted for more than 6 years and therefore maps my specialization. Especially the current software projects within the cloud native landscape provide professional tools that help me to modernize the IT industry.

WORK EXPERIENCE

Cloud Architect & DevOps Engineer
Avaloq Bank

2022 - NOW

As responsible engineer and architect the first milestone was the open-shift migration of the Avaloq integration layer followed by the expansion of the landscape with various new services. Integration with 3Scale API management over monitoring and logging makes this project a big experience in an enterprise environment.

Responsibilities in project:

- Implementing and integrating advanced GitOps deployment strategies
- Extending existing Spring Boot microservices and Angular applications
- Creating new microservices utilizing REST and Kafka for both synchronous and asynchronous operations
- Creating and managing Kafka-Connect solutions using the Strimzi operator
- Automating various tasks with Ansible, including custom Ansible module development
- Monitoring of various microservices in a heterogeneous landscape on a large scale with Prometheus and Splunk
- Utilizing Helm, Kustomize, CDK8S for manifests generation and Pulumi & ArgoCD for deployments

Technologies include:

- **Languages:** Java, Kotlin, Bash, Python, TypeScript, HTML and (S)CSS
- **Frameworks:** Spring Boot, Angular, REST, Kafka, Kafka-Connect
- **Tools:** OpenShift 3/4, 3Scale, Helm, Kustomization, CDK8S, Pulumi, ArgoCD, Ansible, Prometheus Operator, Splunk, Dynatrace, (Co)Lima, K6, GitLab, Tekton

EDUCATION

Since 2024

Udemy Instructor

Tekton - The quick start

As an instructor for Tekton CI/CD, I am passionate about sharing my self-educated expertise gained from enterprise projects. My goal is to empower the community by disseminating knowledge on technologies and assisting individuals in their learning journey.

2016 - 2020

B. Sc. - IT-Security

University of Darmstadt

I pursued a dual study program, earning a Bachelor of Science in IT Security. This program combined academic studies with practical work experience at an IT consulting company. During my studies, I gained in-depth knowledge of IT security principles, software engineering, and automation techniques. Concurrently, I applied this knowledge in real-world scenarios by working in blocks and during holidays at the consulting company, practicing what I had learned with actual customers.

2014 - 2016

State certified technician

Technical College Braunschweig

I completed a two-year full-time education program as a State-Certified Technician in Computer Science. During this program, I gained advanced skills in programming, focusing on technical systems such as controllers and bus systems. I practiced software design methods and applied them in various projects, including a team-based project conducted outside the school setting. My studies also included operating systems and virtualization environments.

Cloud Architect & DevOps Engineer Avaloq Bank

2022

As the engineer and architect for the Enterprise Service Bus (ESB) integration with Avaloq using MuleSoft, the initial step in this project is to guide the bank towards adopting a cloud-native approach.

Responsibilities in project:

- Developing and containerizing Spring Boot and Angular applications
- Defining and establishing Continuous Integration and Delivery pipelines
- Engaging in requirements engineering, translating business needs into technical solutions
- Creation and communication of proposed architectures, along with documenting architectural decisions and implementation guidelines
- Defining and integrating automated testing of Spring Boot and Angular applications using frameworks like JUnit and Jasmine

Technologies include:

- **Languages:** Java, Bash, TypeScript, HTML and (S)CSS
- **Frameworks:** Spring Boot, Camunda, Angular
- **Tools:** Docker, Podman, Kubernetes, OpenShift, Jenkins, Dynatrace
- **Architectures:** Microservice architecture, OAuth2, JWT, GitLab

Full Stack Developer Swiss Governance

2020 - 2022

As an engineer and full stack developer for a public-facing job portal maintained by a Swiss government institution, the main tasks involved maintaining the portal and adding new features.

Responsibilities in project:

- Maintaining and developing features for Angular applications and Spring Boot services
- Simplifying and improving Travis CI build pipelines
- Creating and maintaining Ansible playbooks for legacy service deployments
- Integrating monitoring solutions with Prometheus, Grafana, and Jaeger
- Preparing and migrating microservices to OpenShift 4 including GitOps practices for automated deployments
- Creating comprehensive Kubernetes/OpenShift trainings for developers and stakeholders

Technologies include:

- **Languages:** Java, Bash, TypeScript, HTML and (S)CSS
- **Frameworks:** Spring Boot, Spring Cloud, Spring Batch, Angular
- **Tools:** Ansible, Docker, Podman, Kubernetes, OpenShift, Jenkins, Travis CI, Tekton, JHipster, Helm, Kustomize, ArgoCD, Tekton
- **Architectures:** Microservice architecture, OAuth2, JWT, JHipster

2010 - 2013

IT specialist - system integration

Company GEKA mbh

As an entry into the IT career, I completed an apprenticeship as an IT Specialist in System Integration. During this time, I worked as a system administrator for Windows domains and was responsible for electronic installations in the data center. My role involved managing both hardware and software components, with a significant focus on automation processes.

CERTIFICATIONS

Certified Kubernetes Security Specialist

Earners of this designation are accomplished Kubernetes practitioners.

Hands-on exam demonstrated the requisite abilities to secure container-based applications and Kubernetes platforms during build, deployment and runtime, and are qualified to perform these tasks in a professional setting.

Certified Kubernetes Administrator

Holders of this designation demonstrated the skills, knowledge and competencies to perform the responsibilities of a Kubernetes Administrator.

That means demonstrating in hands-on scenarios in Application Lifecycle Management, Installation, Configuration and Validation

Core Concepts, Networking, Scheduling, Security, Cluster Maintenance, Logging Monitoring, Storage, and Troubleshooting.

Full Stack Engineer Automobile Manufacturer

2019 - 2020

Responsible full stack engineer for the migration and enhancement of the dealer portal for internal employees and partners of an automobile manufacturer. This project involved various backend and frontend development tasks, as well as infrastructure provisioning and automation.

Responsibilities in project:

- Developed and maintained the Magnolia CMS backend using Java
- Created and updated frontend components using JavaScript, jQuery, HTML and (S)CSS
- Provisioned and managed the application stack via Ansible
- Designed and implemented PostgreSQL databases to support application data requirements
- Conducted requirements engineering sessions to gather and document business needs
- Utilized Confluence for documentation and Jira for issue tracking and project management
- Presented project status and updates to stakeholders in regular status presentations
- Implemented customer's CI catalog styles in SCSS to ensure brand consistency
- Developed and executed data migration strategies to transition from legacy systems

Technologies include:

- **Languages:** Java, JavaScript, HTML and (S)CSS
- **Frameworks:** Magnolia, REST
- **Tools:** IntelliJ, Docker, Ansible, jQuery, PostgreSQL, Confluence, Jira

Systems Engineer Consultant Various Clients

2016 - 2019

Systems engineer responsible for automating datacenter infrastructure for various client projects. While these projects primarily focused on software, they also involved hands-on work in the datacenter to meet the growing needs of customers.

Responsibilities in project:

- Performed hardware maintenance for on-premises datacenters, including cable management, storage replacements, and hardware extensions
- Provisioned datacenter infrastructure using Ansible for automation
- Maintained various Linux systems, including RHEL, CentOS, Debian, and FreeBSD
- Managed and updated software products such as Nextcloud and single sign-on (SSO) components
- Assisted in migrating from Hyper-V to the RHEL virtualization platform
- Supported the migration from the RHEL virtualization platform to VMware

Certified Kubernetes Application Developer

Earners of this designation demonstrated the skills, knowledge and competencies to perform the responsibilities of a Kubernetes Application Developer. Earners are able to define application resources and use core primitives to build, monitor, and troubleshoot scalable applications and tools in Kubernetes. The skills and knowledge demonstrated by earners include Core Concepts, Configuration, Multi-Container Pods, Observability, Pod Design, Services and Networking, State Persistence.

Kubernetes and Cloud Native Associate

Basic knowledge of Kubernetes and cloud-native technologies, including how to deploy an application using basic kubectl commands, the architecture of Kubernetes, understanding the cloud-native landscape and projects (storage, networking, GitOps, service mesh), and understanding the principles of cloud-native security was acknowledged by successful passing.

Kubernetes and Cloud Native Security Associate

Participants holding this badge have demonstrated an understanding of the baseline security configuration of Kubernetes clusters to meet compliance objectives, including the ability to harden security controls, test and monitor the security, and participate in assessing security threats and vulnerabilities.

- Configured and maintained pfSense firewalls, including regular updates
- Conducted analyses of diverse IDS solutions, comparing open source and commercial products
- Conducted proof of concept (PoC) and implemented Snort IDS/IPS Firewall in datacenters
- Implemented runtime security for Kubernetes on-premises
- Provisioned vulnerable applications for security test cases
- Integrated and maintained ELK (Elasticsearch, Kibana, Logstash) stacks
- Integrated and maintained Graylog for log management
- Updated customer-facing applications and managed infrastructure and user accounts using Ansible

Technologies include:

- **Languages:** Bash, Golang, Python, TypeScript
- **Frameworks:** ELK (Elasticsearch, Kibana, Logstash)
- **Tools:** IntelliJ, Docker, Falco, Sysdig, Kubernetes, Ansible, Snort, Nextcloud, Graylog

PCA: Prometheus Certified Associate

Earners of this designation demonstrated an understanding of best practices for monitoring cloud native applications and infrastructure using Prometheus, including how to use observability data to improve application performance, troubleshoot system implementations, and feed that data into other systems.

ICA: Istio Certified Associate

Earners of this designation demonstrated a solid understanding of Istio principles, terminology, and best practices in order to set up Istio.

CAPA: Certified Argo Project Associate

Earners of this designation demonstrated a solid understanding of the Argo Project ecosystem, terminology, and best practices for each tool and how it relates to common DevOps, GitOps, Platform Engineering, and related practices.

CGOA: Certified GitOps Associate

Earners of this designation demonstrated comprehension of GitOps patterns including when and why to use them. They also understand the precautions to take around tooling for manifest format and packaging, state store systems, and reconciliation engines to adhere to and extend GitOps principles.